Stacked Channel Transistors with 2D Materials

An Integration Perspective

Presented for EE6346:Advanced CMOS Devices and Technology
Mirudhula J
EE23B046

20/05/2026

Outline of Presentation

Background & Motivation

Single NS Devices

Stacked NS Process

Electrical Results

CFET Structure

Key Challenges

Conclusions

The Transistor Scaling Challenge

- Moore's Law demands smaller, faster, lower-power transistors each generation
- FinFET technology is approaching fundamental physical limits below **3 nm**
- Nanosheet / Gate-All-Around (GAA) FETs are the next architecture
  - Gate wraps *completely* around a flat sheet channel
  - Far better electrostatic control than FinFET
- Core problem: what replaces silicon as the channel material?
  - Si degrades badly below ~3–4 nm thickness
  - Rough interfaces, high defect density, poor mobility
- **Answer:** 2D Transition Metal Dichalcogenides (TMDs)

Why 2D Materials (TMDs)?

Structure: MX_2 formula

- M = Mo or W   X = S or Se
- Monolayer thickness: **only ∼0.7 nm** (3 atoms!)
- Natural bandgap (unlike graphene) — works as a transistor
- No dangling bonds at surface — clean interface

Two key materials:

- **MoS_2** — NMOS (n-type) candidate
- **WSe_2** — PMOS (p-type) candidate

Electrostatic Advantage

Ultra-thin body enables extreme gate control ⇒ gate length scaling to **sub-10 nm**

Current Challenge

Mobility and drive current still lower than silicon — need a structural solution

Why Stack the Nanosheets?

- One TMD sheet = too little current for logic circuits
- **Solution:** Stack 2, 3, or 4 sheets — each is an independent channel, all sharing one gate
- TCAD modeling (Fig. 1 of paper):
  - I_{cell} grows *faster* than C_{cell} with more sheet; better speed or lower power
- Best single-NS result: $\sim$400,μA/μm, SS $\sim$86,mV/dec (1L-MoS_2)
- First electrical demonstration of a stacked NS FET with a 2D TMD channel

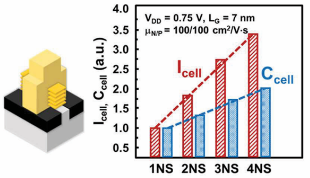


Fig: Sketch of a stacked TMD NS channel transistor. Normalized Icell and Ccell performance from TCAD.

Single NS: Gate Dielectric Study on MoS_2

Short-loop flow (no channel release) used to compare interfacial layers (IL):

Property	**HfO_2 IL (50°C ALD)**	**$SiON_x$ IL (150°C PECVD)**
IMAX	$\sim$100 μA/μm **(higher)**	Lower
On/Off Ratio	$\sim 10^9$ **(9 orders)**	Lower
SS (mV/dec)	$\sim$190	Better (lower)
Hysteresis	More	**Nearly zero**
Variability	Higher	**Lower**

- Higher-temperature $SiON_x$ chemically damages MoS_2 $\Rightarrow$ lower current
- But creates cleaner interface $\Rightarrow$ fewer charge traps, less hysteresis

Single NS: WSe_2 — The Mechanical Challenge

Problem:

- WSe_2 has weaker mechanical strength than MoS_2
- During wet cleaning and etch steps, sheets tear apart
- Low yield of suspended WSe_2 nanosheets

Solution: Protective dielectric sandwich

- Deposit thin dielectric above and below
- Acts as mechanical support; Increases etch selectivity
- Widens process window for high-k deposition

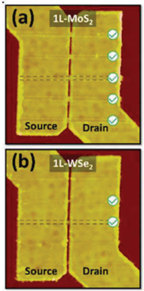
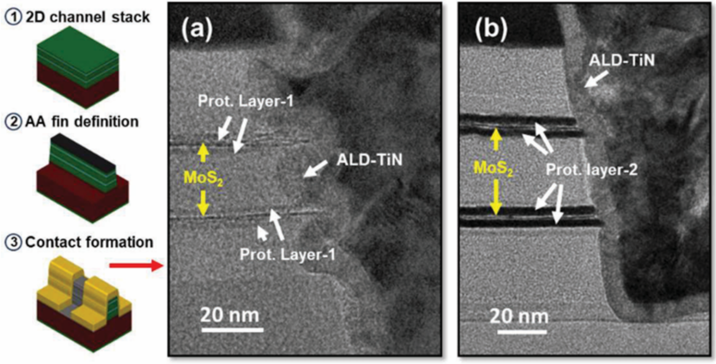


Fig: Example stacked nanosheet structure

Key Insight

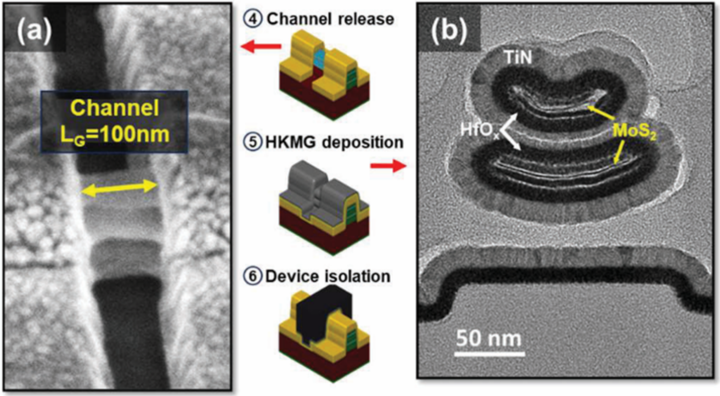
Same protective dielectric concept will be essential for stacking

Stacked NS: Contact Formation



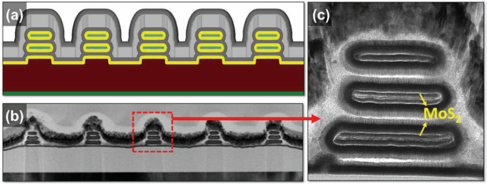
- ALD-TiN contact deposition
- Atomic Layer Etching (ALE) removes TiN from channel region
- Sacrificial layer etch $\Rightarrow$ NS release (sheets suspended)
- HKMG deposition

Stacked NS: Solving the "Smiling" Problem

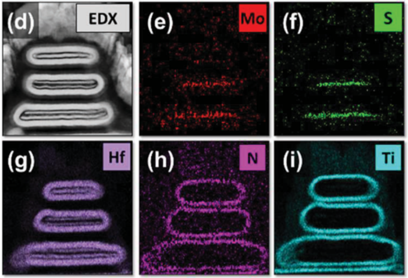


- HfO_x/TiN HKMG deposition induced uniaxial stress, causing nanosheets to bend upward ("smiling")
- Bent sheets weakened contact anchoring and reduced fabrication yield
- ALD parameters were optimized to balance film stress
- Final process achieved flat nanosheets with fully conformal HKMG

HKMG Deposition - After optimization



(a) Process software model and (b) TEM cross-section of 2-stacked NS FET with 5 fingers. (c) Zoom-in TEM image



(d) - (i) EDX mappings of stacked 1L-MoS2 NS channel with HKMG.

First Electrical Results: 2-Stacked MoS_2 NS FET

Device details:

- 2-stacked 1L-MoS_2 NS FET
- $L_G = 100\,\text{nm}$ to $250\,\text{nm}$
- Single-finger ($1\times$) and multi-finger ($5\times$)

Median results at $V_{DS} = 1\,\text{V}$:

IMAX	$\sim 4\,\mu\text{A/sheet}$
On/Off ratio	$\sim 10^5$
$V_{\text{TH,lin}}$	$\sim 0\,\text{V}$
SS	$\sim 220\,\text{mV/dec}$

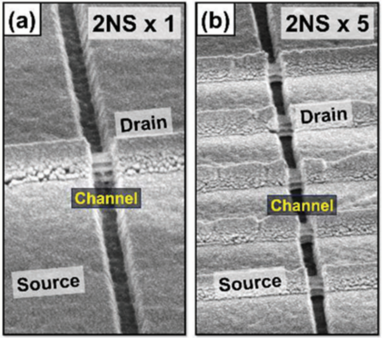
Why is IMAX so low?

- **Edge-only contact** (not C-type)
- **TiN** contact metal (not Sb)
- Gate works fine — channel is not the bottleneck

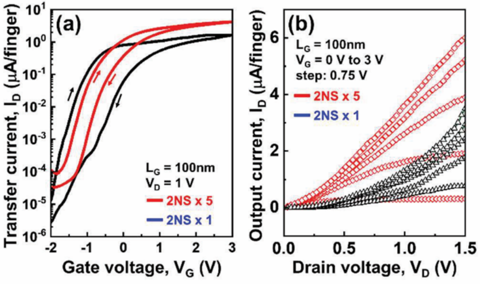
Multi-finger advantage

$5\times$ fingers give: higher IMAX, lower hysteresis, better output linearity $\Rightarrow$ contact quality issue at edges in $1\times$

Figures supporting the setup and the performance



2-stacked NS after channel release for (a) 1-finger and (b) 5-finger channels.



(a) Transfer and (b) Output characteristics for 2-stacked NS with 1 and 5 fingers respectively.

Gate Length Scaling: Contact Resistance is the Bottleneck

SEM observation:

- Sheets remain suspended **without sagging** up to $L_G = 250\,\text{nm}$
- Enhanced stiffness from the protective dielectric sandwich
- Demonstrates mechanical viability of the structure

Electrical observation:

- As L_G decreases from $250\,\text{nm} \rightarrow 100\,\text{nm}$, ION barely increases
- In a channel-limited device, shorter L_G = much higher ION
- Here: no strong ION improvement $\Rightarrow$ contact resistance dominates

Physical Interpretation

Total $R = R_{\text{contact}} + R_{\text{channel}}$

$R_{\text{contact}} \gg R_{\text{channel}}$

$\therefore$ shortening the channel changes nothing — the bottleneck is not there

Path forward

C-type Sb contacts $\Rightarrow$ expected dramatic IMAX improvement

CFET Concept

What is CFET?

- Complementary FET: PMOS stacked *directly on top of* NMOS
- Same horizontal footprint as a single transistor
- Ultimate transistor density scaling — the industry's long-term goal

For 2D materials, the natural pairing is:

- **MoS_2 (bottom)** — NMOS
- **WSe_2 (top)** — PMOS

This work: 4 channels total

Stacking order (bottom → top)

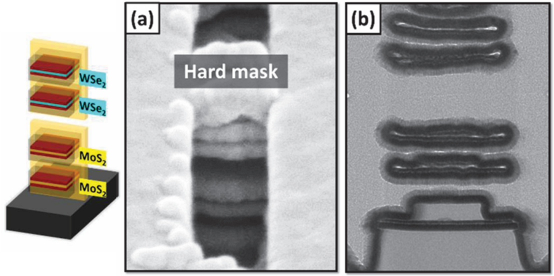
MoS_2 sheet 1
SAC layer (sacrificial)
MoS_2 sheet 2 + protective dielectric
MDI (mid-dielectric isolation)
WSe_2 sheet 3 + protective dielectric
SAC layer
WSe_2 sheet 4

CFET: Fabrication and Verification

After selective SAC etch (NS release):

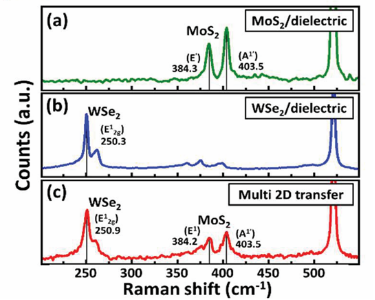
- SEM: all 4 sheets successfully suspended — no tearing
- Critical: protective dielectric gives the 0.7 nm sheets enough stiffness to survive

After HKMG deposition:



(a) Tilted SEM and (b) TEM images of stacked WSe2 and MoS2 NS channels after HKMG deposition.

CFET: Raman Verification



Raman spectroscopy:

- MoS_2 peaks at ~384 and ~404 cm^{-1} intact
- WSe_2 peak at ~251 cm^{-1} intact
- **Monolayer crystal structure preserved**

Why Raman matters

If the TMD were damaged (strained, cracked, or multilayer-fused), these Raman peaks would shift or split. The fact they don't confirms the 2D material survived all the processing steps intact.

Key Process Innovations and Open Challenges

Innovations (solved in this paper):

- Protective dielectric sandwich
  $\Rightarrow$ survives etch & cleaning steps
- ALD stress optimization
  $\Rightarrow$ flat sheets, no "smiling"
- ALE for TiN removal from channel
  $\Rightarrow$ atomic-precision etch
- SAC layer integration for CFET
  $\Rightarrow$ 4-channel release demonstrated

Still open (future work needed):

- C-type contact not yet achieved in full flow
  need conformal ALD Sb deposition
- SS still $\sim$220 mV/dec (ideal: 60)
  interface traps from HfO_2 process
- Sub-100 nm gate length
  not yet demonstrated with 2D channel
- Separate gate control of NMOS/PMOS in CFET
  shown here as single-gate only
- Etch selectivity chemistry for 2D materials
  deeper fundamental work required

Conclusions

World firsts in this paper:

- **First** electrical stacked NS FET with 2D channel
- **First** 4-channel stacked CFET-like structure
  $2\times MoS_2 + 2\times WSe_2$, all released and gate-wrapped

Key numbers:

- IMAX $\sim 4\,\mu A$/sheet, On/Off $\sim 10^5$ (contact-limited)
- NS mechanically stable up to $L_G = 250\,\text{nm}$
- Raman confirms monolayer integrity after full process

Proves that 2D materials can be integrated into **production-compatible** nanosheet process flows — the fabrication challenge is solvable

Immediate next steps

- Conformal ALD Sb for C-type contacts
- Sub-100 nm L_G with 2D channel
- Independent gate control in CFET

Thank You

Questions?